MUSHFIQUE TANZIM MUZTABA

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PROFESSIONAL SUMMARY

Final-year Mechanical Engineering student at The Hong Kong Polytechnic University skilled in mechanical design (SOLIDWORKS, Finite Element Analysis), robotics (ROS2, SLAM, Nav2), AI/computer vision (YOLO, OpenCV), and engineering workflow automation (Power Platform), delivering custom tools that reduced project turnaround times by up to 30%. Seeking graduate roles in engineering and technology.

EDUCATION

The Hong Kong Polytechnic University **Hong Kong**

Bachelor of Engineering (Honours) in Mechanical Engineering *Graduation Date: May 2026*

- **Accolades:** The Hong Kong Polytechnic University Entry Scholarship (Academic)
- **Relevant Coursework:** Manufacturing Fundamentals, Numerical Methods for Engineers, Fluid Mechanics, Engineering Thermodynamics, Linear Systems and Control, Fundamentals of Robotics

WORK EXPERIENCE

MangDang Technology Co., Limited **Hong Kong**

Robotics Intern (Part-time) *March 2026*

- Migrated quadruped robot software stack from ROS2 Humble to Jazzy, including simulator transition from Gazebo Classic to Gazebo Harmonic and full Ubuntu 24.04 compatibility.
- Re-engineered the CHAMP quadruped contact sensor node from Gazebo transport APIs to GZ Harmonic, improving gait control accuracy.
- Validated autonomous navigation by testing SLAM (Cartographer, SLAM Toolbox), Nav2, and teleoperation on both simulator and Raspberry Pi hardware.

CLP Power Hong Kong **Hong Kong**

Sandwich Intern *July 2024 – June 2025*

Project Execution Team (PET)

- Built a Power Apps 'Lessons Learned' portal for 100+ members across 4 business groups to capture and share project insights, with planned company-wide rollout.
- Created multi-stage approval workflows with engagement features (likes, leaderboards) in Power Automate, cutting feedback turnaround time by 30%.
- Completed Project Management Academy training in CLP's project governance system.

Hong Kong Battery Energy Storage System (HKBESS)

- Developed a Power Apps Management-of-Change tool for the Hung Shui Kiu "A" substation, centralizing engineering change requests.
- Engineered multi-level approval routing in Power Automate for HKBESS, reducing cycle time by 40% with real-time Teams notifications.
- Documented 15+ operational risks for the battery storage system, informing project safety protocols.

Thai German Graduate School of Engineering (TGGS) **Bangkok, Thailand**

Research Intern *August 2023*

- Designed a motorbike crash-test dolly in SOLIDWORKS from concept sketches to production-ready assemblies.
- Performed iterative FEA under simulated crash loads, optimizing geometry for target safety factors and material efficiency.
- Engineered an energy-absorbing crash-cushion system, reducing peak impact forces and improving test repeatability.

PROJECT EXPERIENCE

Autonomous Indoor Thermal Inspection Robot **January 2026 – May 2026**

- Developed an autonomous indoor inspection robot using ROS 2, SLAM Toolbox, and frontier-based exploration with LiDAR, RGB, and thermal sensors for non-contact anomaly detection (>35°C).
- Built a multi-layer perception pipeline combining thermal hotspot detection, object detection (YOLO11s), and Gemini 2.5 Flash multimodal analysis on a Raspberry Pi 5 with distributed processing via rosbridge.
- Implemented a natural-language control interface (Gemini 2.5 Flash) and operator dashboard (React) with live SLAM map, camera feeds, and thermal visualization.

Autonomous Vision-Guided Targeting System **November 2025**

- Developed an autonomous targeting system in Python/OpenCV on Raspberry Pi, achieving 90% detection accuracy under variable lighting via HSV colour thresholds.
- Implemented PID controller (Kp=0.12, Ki=0.005, Kd=0.25) with frame smoothing and dead zones for precise 2-axis servo control via I2C PWM.
- Integrated trajectory compensation and GPIO-controlled firing logic for reliable target acquisition at 1 m.

AI Multi-Model Object Detection System **October 2025**

- Trained a YOLOv11n model to 90–94% accuracy on a custom 100+ image dataset (PyTorch, Roboflow).
- Combined YOLO detection with HSV colour classification for multi-attribute identification (82–99% confidence across categories).
- Built a complete ML pipeline from data labelling to model deployment using Python and OpenCV.

Personal Portfolio Website **August 2025**

- Developed a responsive engineering portfolio using React 19, TypeScript, and Vite to showcase robotics and mechanical design projects.
- Engineered interactive features including scroll-based animations (Framer Motion) and a custom Canvas particle background, with smooth scrolling and full mobile responsiveness.

PolyU E Formula Racing Team **September 2023 – May 2024**

- Optimised suspension geometry using Lotus Suspension Analysis to evaluate handling characteristics for the team's electric formula vehicle.
- Designed tie rods, steering column, and steering linkages in SOLIDWORKS, ensuring manufacturability and assembly fit.

Automated Material-Handling Trolley **September 2023**

- Designed and manufactured an automated industrial trolley in SOLIDWORKS with motorized drive and steering mechanisms.
- Implemented autonomous loading/unloading, straight-line tracking, and 90° turning within a fixed-footprint environment.

TECHNICAL SKILLS

- **CAD and Mechanical Design:** SOLIDWORKS (modelling, FEA, assemblies), AutoCAD
- **Programming:** Python, C++, MATLAB
- **Robotics:** ROS2 (Humble, Jazzy), Gazebo, SLAM Toolbox, Cartographer, Nav2
- **AI and Computer Vision:** PyTorch, YOLOv11, OpenCV
- **Embedded Systems:** Raspberry Pi, STM32, I2C, SPI, GPIO
- **Engineering Tools:** Microsoft Power Apps, Power Automate, Git, Office 365
- **Web Development:** React, TypeScript, Tailwind CSS, Vite, Framer Motion

LICENSES AND CERTIFICATIONS

- Hong Kong Green Card (Construction Industry Safety Training) *Valid until March 2028*
- Hong Kong Blue Card (Shipboard Cargo Handling Basic Safety Training) *Valid until January 2028*

LinkedIn Learning Certificates: SOLIDWORKS: Advanced Engineering Drawings, SOLIDWORKS: Advanced Simulation, SOLIDWORKS: Modelling Gears, Learning AutoCAD 2024.

LEADERSHIP AND ACTIVITIES

PolyU ENGL English Debate Club **Hong Kong**
Creative Designer *August 2024 – June 2025*

- Designed posters, social media graphics, and event merchandise, strengthening brand engagement for 500+ members.

PolyU International Student Association **Hong Kong**
Creative Designer and Social Media Manager *July 2023 – June 2024*

- Created promotional graphics and videos garnering 15,000+ views, boosting event participation.

PolyU Student Halls of Residence **Hong Kong**
Event Organizer *November 2022 – June 2025*

- Organized and coordinated inter-hall activities for 120+ participants, fostering community engagement.